is a highly parallel runtime that allows Elixir to seamlessly scale across all CPU cores. This gives it an advantage when it comes to squeezing more performance out of a system for less total cost.

Processes are the foundation of concurrency in Elixir. Processes send and receive messages with other processes, and it's common to run hundreds of thousands of processes in a production system. This is because processes are lightweight, with each having a separate memory allocation and garbage collector.

Processes are easy to work with, but building production systems requires a lot of cumbersome tasks to get right. The `GenServer` module takes care of these tasks for us. GenServer lets us create long-lived processes that maintain state and process messages easily. There's no magic here though—GenServer is built on top of processes, so we get all of the benefits of processes like error isolation and memory independence.

Elixir is highly parallel, but systems usually can't run at full firepower. It's important to control the amount of parallelism in an application so that you don't overwhelm external systems like databases or APIs. Libraries usually handle this pretty well for you, so you have a good starting place to make sure your application is stable from day one. In fact, existing libraries do so much for you that you may not actually need to worry about GenServers and processes for some time.

The BEAM's process model allows us to build completely different styles of applications than we would otherwise. Instead of traditional single-stack-oriented applications where everything is in one big stack, we can separate our system into smaller subsystems that communicate using message passing. Think of your system as a city that you can expand or shrink easily over time.

This wraps up the first part of this book. The next part of this book is going to see us through a real-world project. You'll learn about the most popular libraries and how they enable you to build scalable applications built on the BEAM's process model.